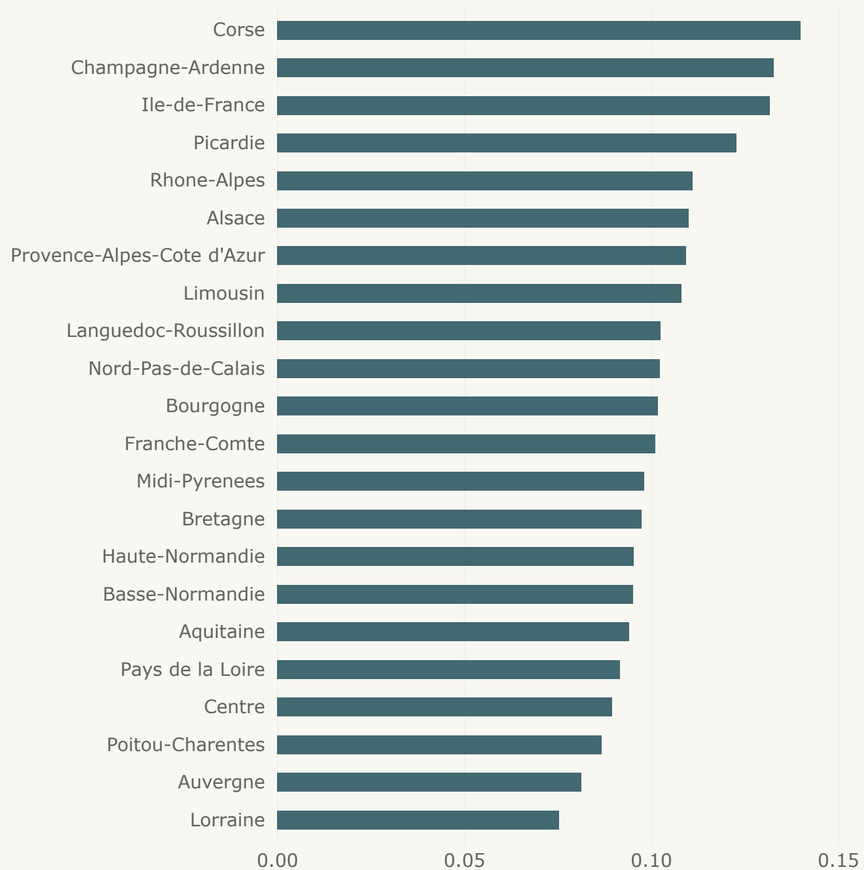
Portfolio Overview

Policies	Policy-Years	Claims	Claims / Policy-Year	Avg Cost per Claims	Exp. Cost / Policy-Year
678.01K	358.36K	36K	0.1006	€ 1.569K	€ 237

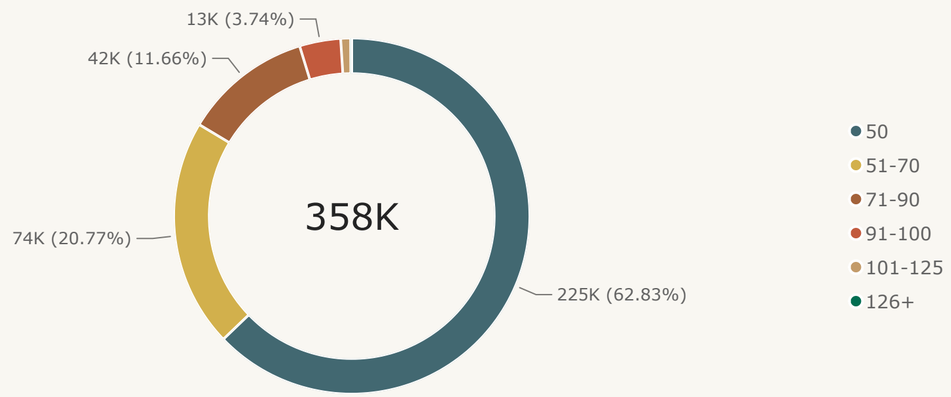
Which Regions claim most often?

Claims per Policy-Year by Region

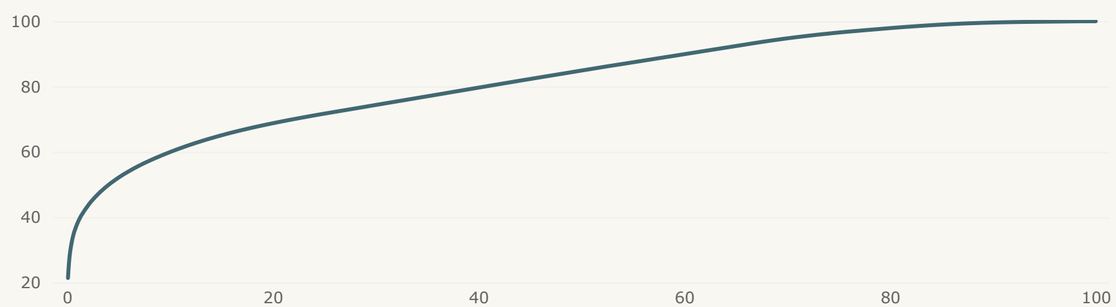


Who do we insure?

Share of insured time (policy-years) by bonus-malus level



% of Total Claim Cost from the Largest x% of Claims



Region

All

Area

All

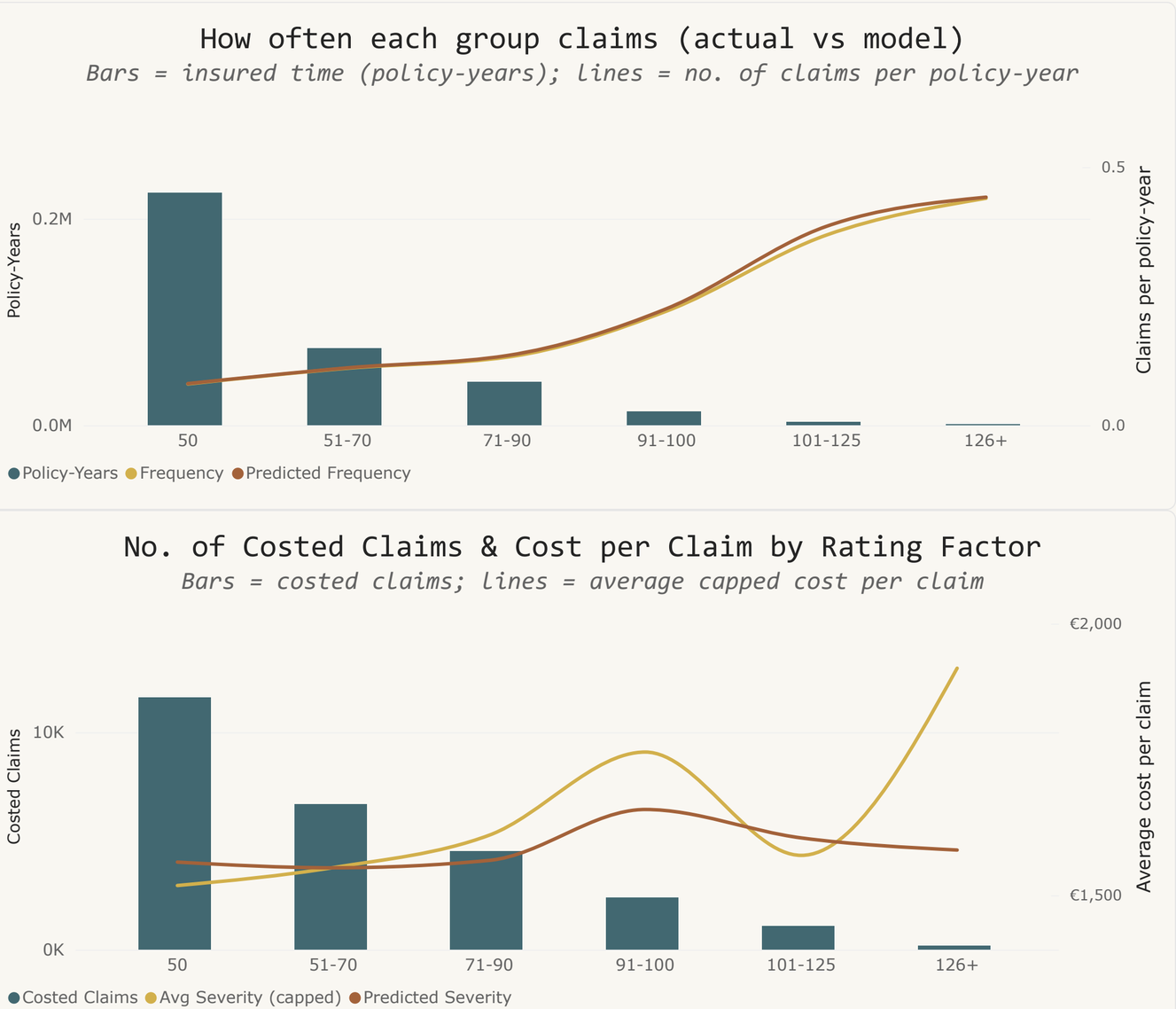
Driver Age

All

Fuel type

All

Risk Drivers



Rating Factor

Region	Vehicle Brand	Vehicle Age
Density Band	Fuel Type	Bonus Malus
Area	Vehicle Power	Driver Age

RegionName

Select all	Aquitaine
Alsace	Auvergne

Matrix Heatmap

Frequency by Driver Age and Bonus Malus

Driver age ↓ / Bonus-malus →	50	51-70	71-90	91-100	101-125	126+
18-20	0.135	0.308	0.090	0.286	0.536	0.489
21-25	0.124	0.055	0.110	0.245	0.378	0.453
26-30	0.052	0.069	0.121	0.207	0.291	0.344
31-40	0.061	0.094	0.126	0.189	0.375	0.386
41-50	0.089	0.138	0.162	0.202	0.358	0.438
51-60	0.082	0.145	0.165	0.207	0.370	0.418
61-70	0.079	0.163	0.180	0.208	0.323	0.552
71+	0.083	0.182	0.208	0.208	0.535	0.771

Model Performance

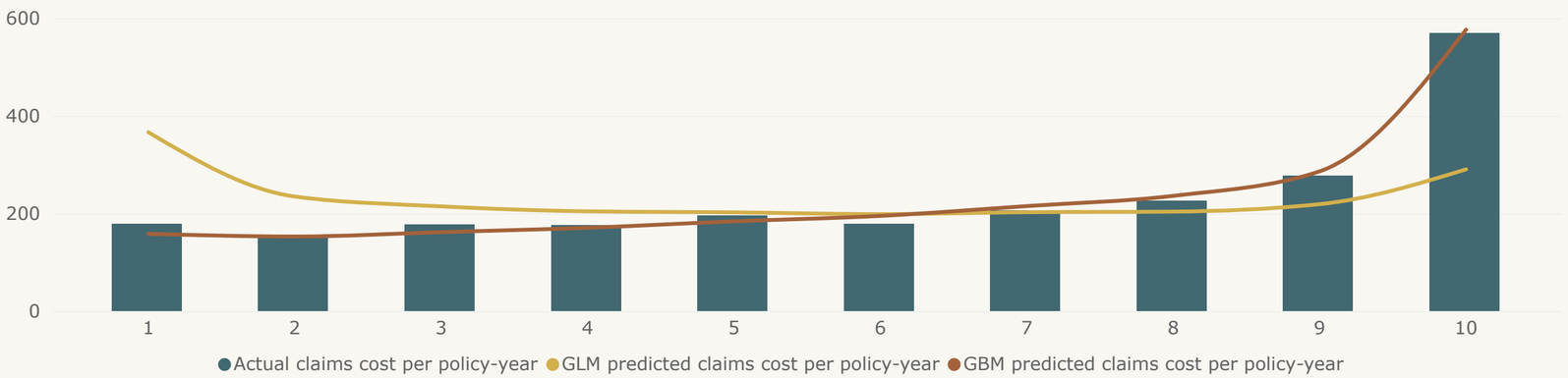
Actual / Predicted Claims Cost	GLM Gini	GBM Gini
0.985	0.30	0.35

Model: Frequency (negative…

Factor: Vehicle Age

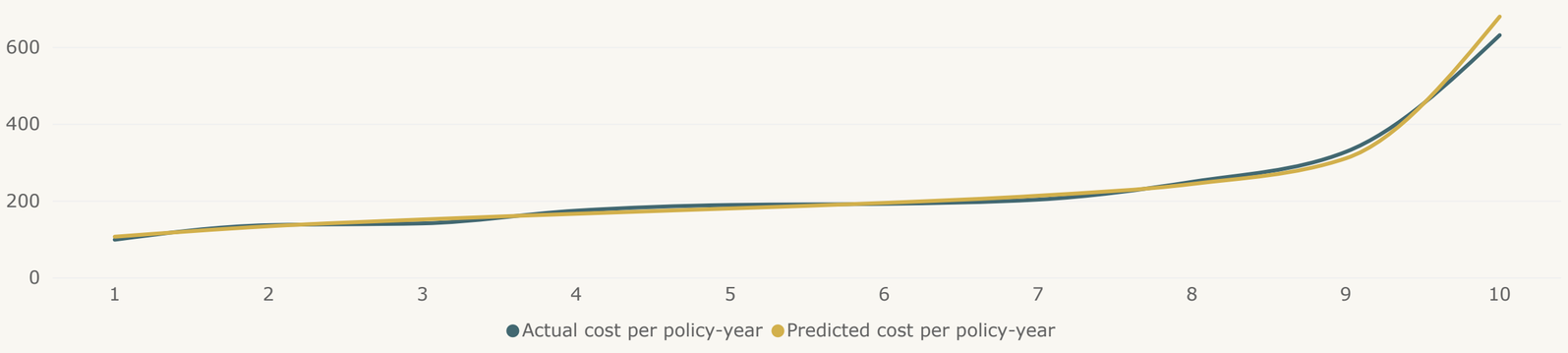
GBM vs GLM Predicted Claims Cost by Decile Groups

Bars = actual claims cost; lines = predicted claims cost

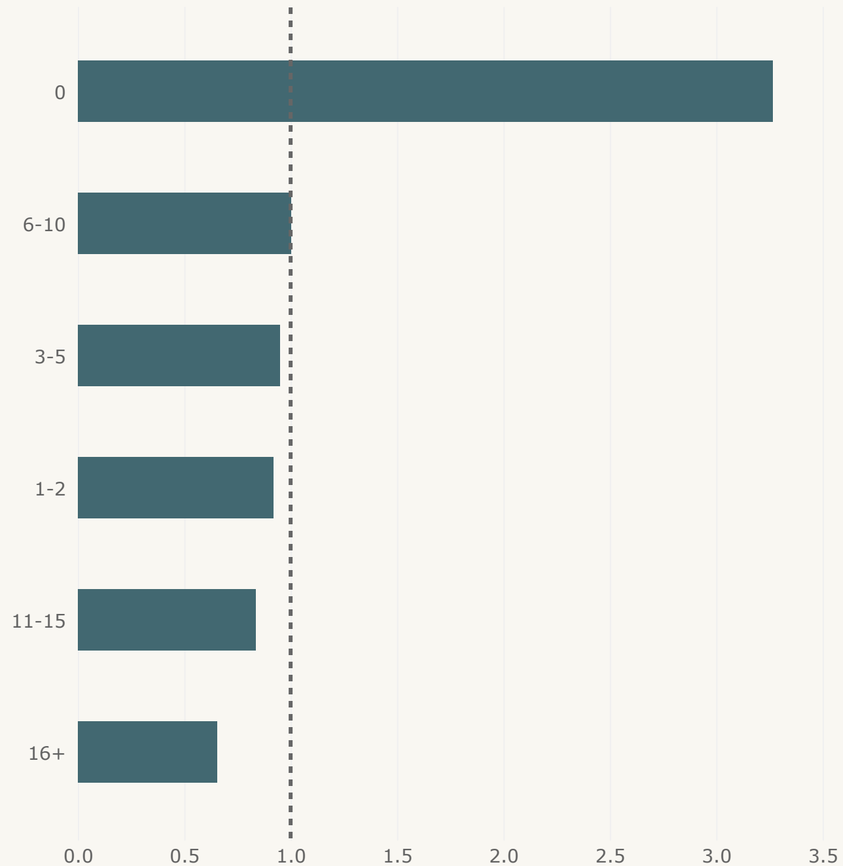


Claims cost per policy-year

Unseen policies sorted into 10 groups (decile) by predicted cost



How the price is built: rating factor multipliers



Pricing Adequacy & Cross-Subsidy

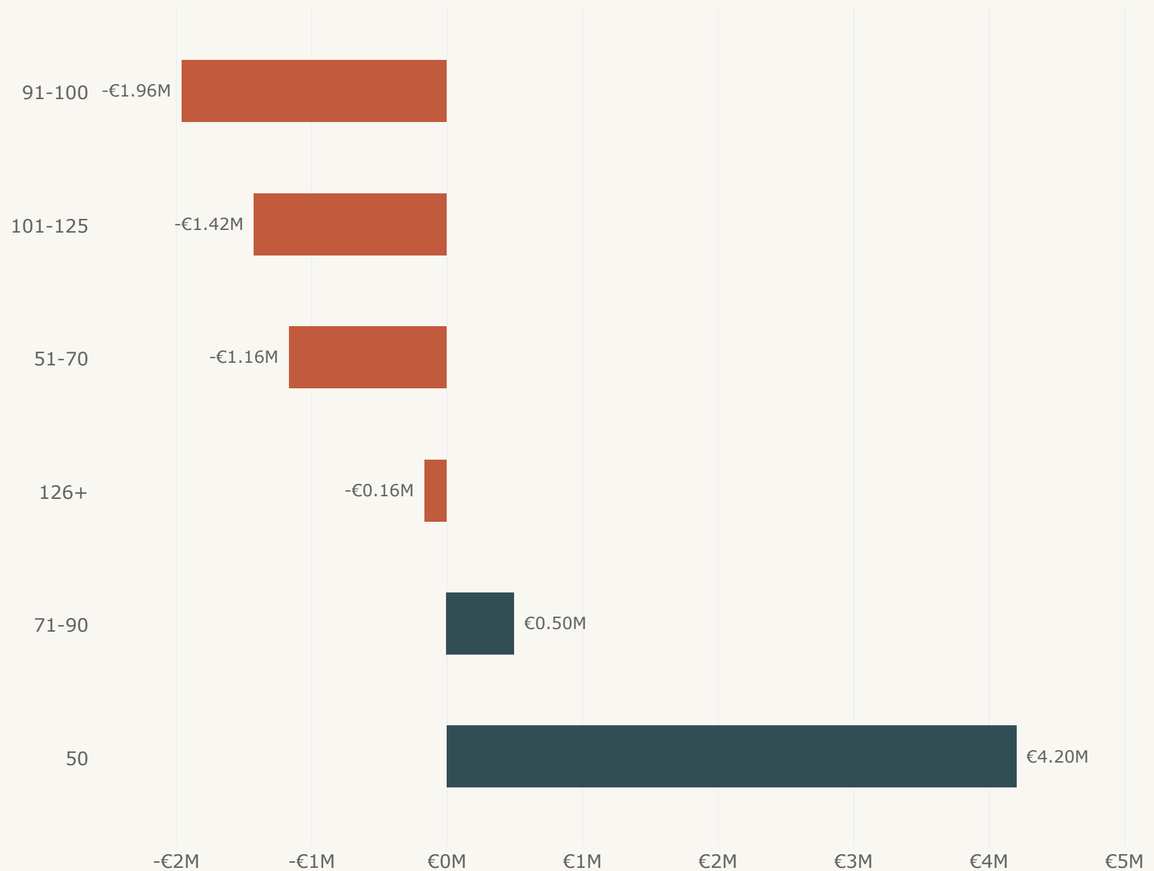
Premium Shifted	% Premium Shifted	% Overcharged >20%	% Undercharged >20%
€15M	17.7%	48.6%	11.3%

Rating Factor: Bonus Malus

Tariff: Bonus-malus only

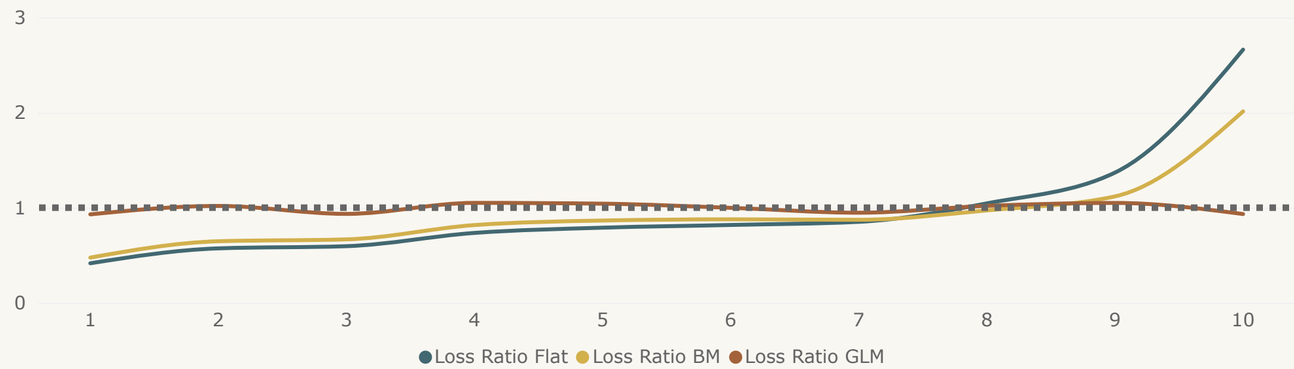
Who Pays for Whom (Premium - Expected Claims Cost per Year)

Orange = subsidised; green = pays for others



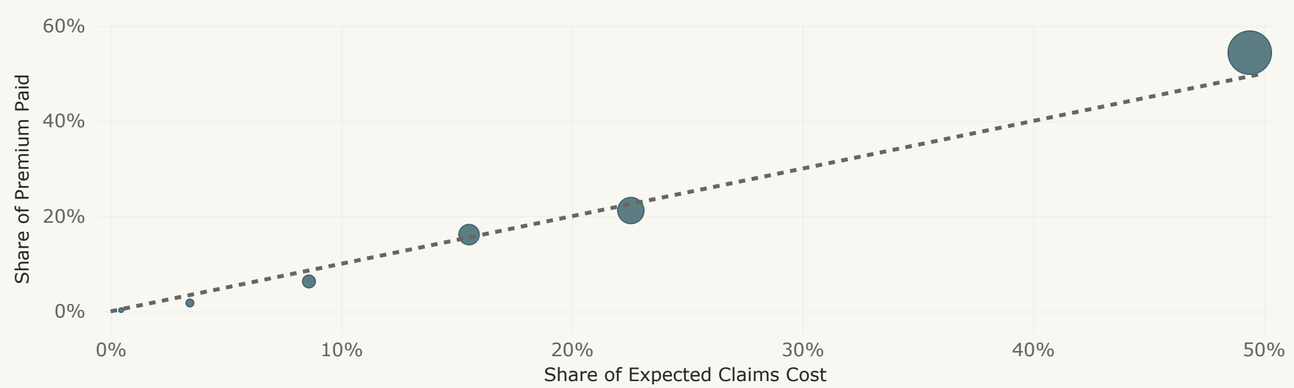
Loss Ratio (claims / premium) on unseen policies by risk decile

1.00 = break-even



Share of premium vs share of expected claim cost

Above the line = pays more than it costs; below = subsidised



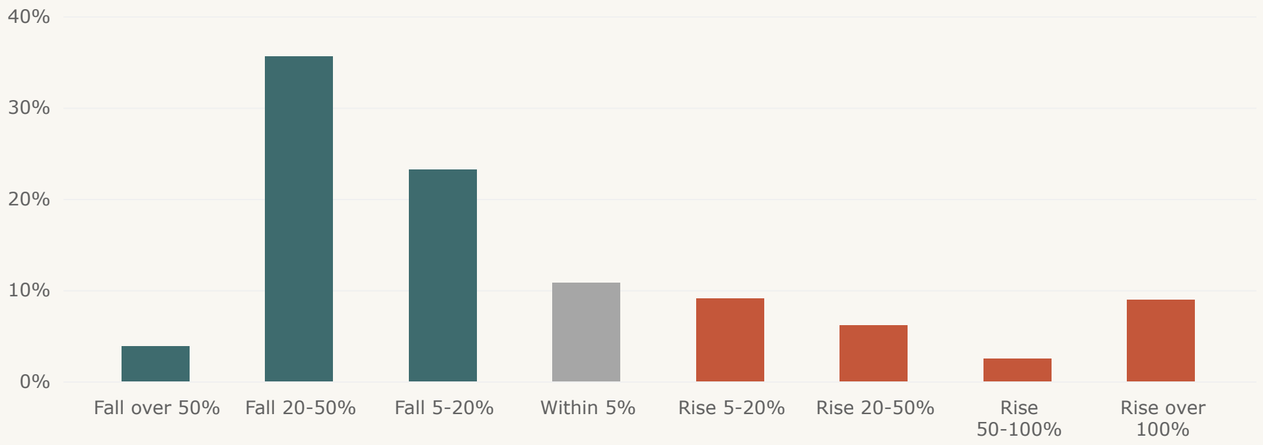
Risk-Based Pricing

% Customers with >20% Decrease	% Customers with >20% Increase	% Customers with >100% Increase
39.4%	17.5%	8.92%

Rating Factor: Bonus Malus

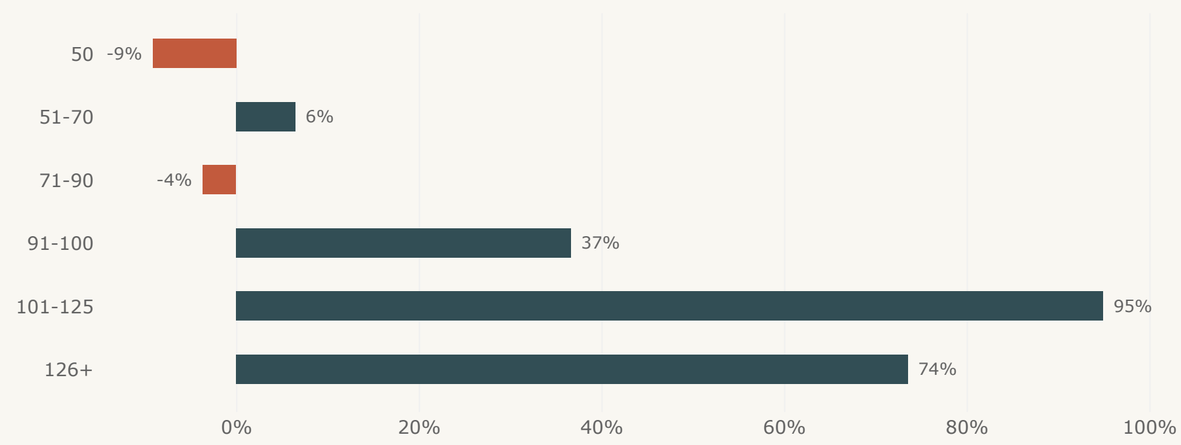
How would prices change: bonus malus -> GLM

Share of customers by size of price change



Which groups would pay more?

Price change (%) from bonus-malus tariff to GLM



What if Table

Tariff	Leave if overcharged by	Policy-years lost (%)	Premium lost (%)	Loss ratio after	Price rise needed (%)
Bonus-malus only	10%	61.50	59.63	1.42	42.37
Bonus-malus only	20%	48.64	47.20	1.29	29.35
Bonus-malus only	30%	37.26	36.14	1.21	20.83
Flat	10%	66.19	66.19	1.64	63.63
Flat	20%	57.01	57.01	1.47	47.23
Flat	30%	46.94	46.94	1.34	34.47